Practical No – 7

Write a python program to read data from a CSV file, perform simple data analysis, and generate basic insights. (Use Pandas is a Python Library)

```
Import pandas as pd;

# 1.Read Program

File_path = “data_sales.csv”

try:

   df = pd.read_csv(file|_path)

   print(“\nCSV Loaded Successfully\n”)

except FileNotFoundError:

    print(“\n File not found. Ensure the CSV is in the same folder.”)

    exit()

print(“Print 5 rows”)

print(df.head())


print(“Data structure”)

print(df.info())


print(“If empty Value Present in the data”)

print(df.isnull().sum())


print(“Stats Summary of the data”)

print(df.describe())
```

OUTPUT:-

```
PS C:\Users\Aishwarya\OneDrive\Documents\Data Analysis> python practicle7_data_an.py

CSV Loaded Successfully

Print 5 rows
   Index      Customer Id First Name                         Company                       Email  Sales
0      1  DD37Cf93aecA6Dc     Sheryl                 Rasmussen Group     zunigavanessa@smith.info   4000
1      2  1Ef7b82A4CAAD10    Preston                     Vega-Gentry              vmata@colon.com   5000
2      3  6F94879bDAfE5a6        Roy                    Murillo-Perry          beckycarr@hogan.com   3040
3      4  5Cef8BFA16c5e3c      Linda  Dominguez, Mcmillan and Donovan  stanleyblackwell@benson.org   2344
4      5  053d585Ab6b3159     Joanna         Martin, Lang and Andrade      colinalvarado@miles.net   4355
Data structure
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 6 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Index        6 non-null      int64
 1   Customer Id  6 non-null      object
 2   First Name   6 non-null      object
 3   Company      6 non-null      object
 4   Email        6 non-null      object
 5   Sales        6 non-null      int64
dtypes: int64(2), object(4)
memory usage: 420.0+ bytes
None
```

```
None
If empty Value Present in the data
Index           0
Customer Id     0
First Name      0
Company         0
Email           0
Sales           0
dtype: int64
Stats Summary of the data
           Index         Sales
count  6.000000      6.000000
mean   3.500000   3493.500000
std    1.870829   1132.430263
min    1.000000   2222.000000
25%    2.250000   2518.000000
50%    3.500000   3520.000000
75%    4.750000   4266.250000
max    6.000000   5000.000000
```

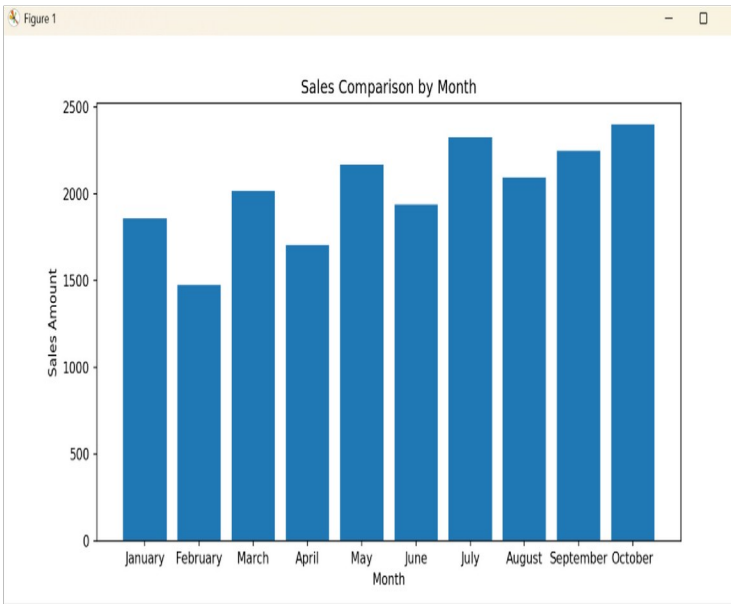
Practical No – 8

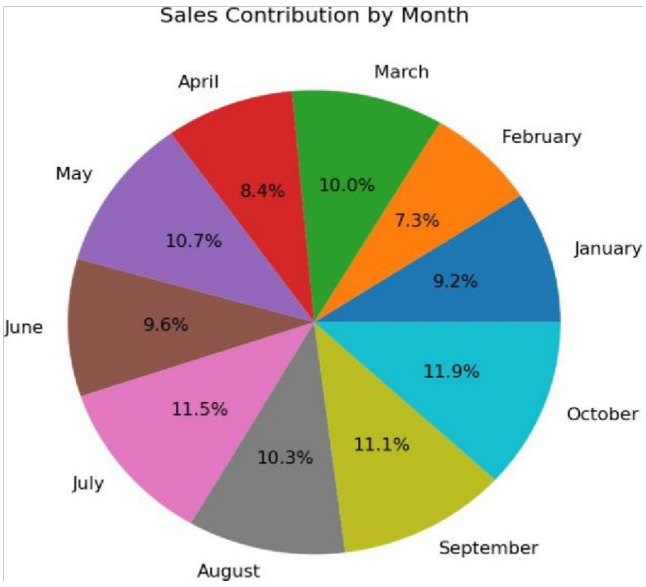
A. Perform data visualization using python on any sales data.

```
import pandas as pd;
import matplotlib.pyplot as plt
file_path = “sale_data.csv”
df = pd.read_csv(file_path)
#print(df)
#Line Plot Month Wise Sales
plt.figure(figsize=(10,5))
plt.plot(df[“Month”],df[“Total_Sales”], marker= “0”)
plt.title(“Monthly Sales Trend”)
plt.xlabel(“Month”)
plt.ylabel(“Sales Amount”)
plt.show()
```

```
#Pie Chart
Plt.figure(figsize=(7,7))
plt.pie(df["Total_Sales"], labels=df["Month"], autopct="%1.1f%%")
plt.title("Sales Contribution by Month")
plt.show()
```

OUTPUT:-





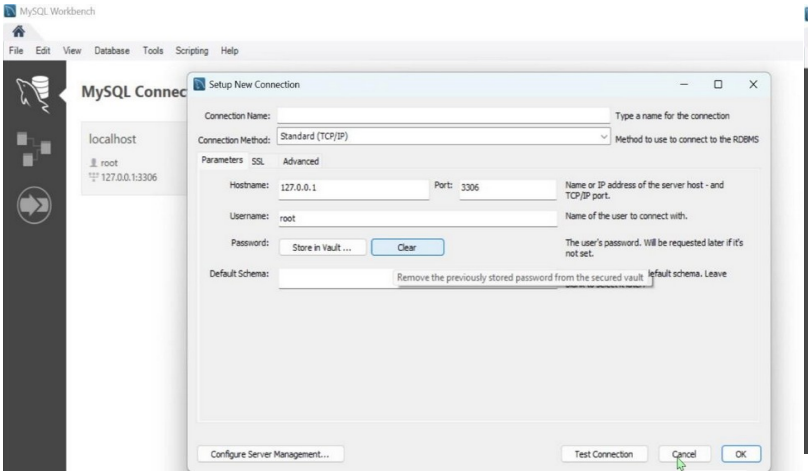
Practical No – 9

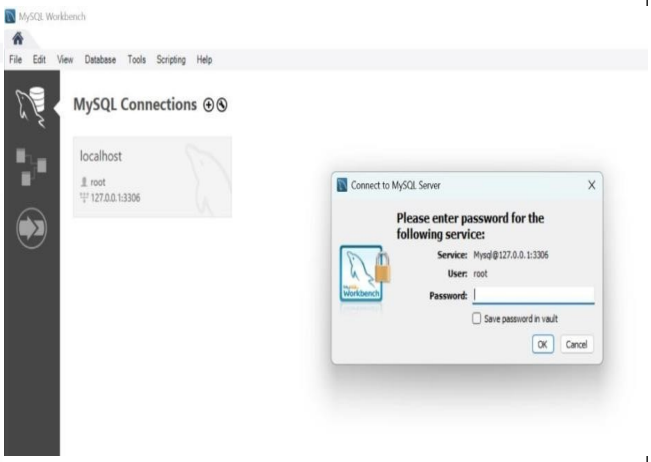
Create the data staging area for the selected database using SQL.

Step 1: Open MySQL Workbench

Launch MySQL Workbench and connect to the **local host** server using:

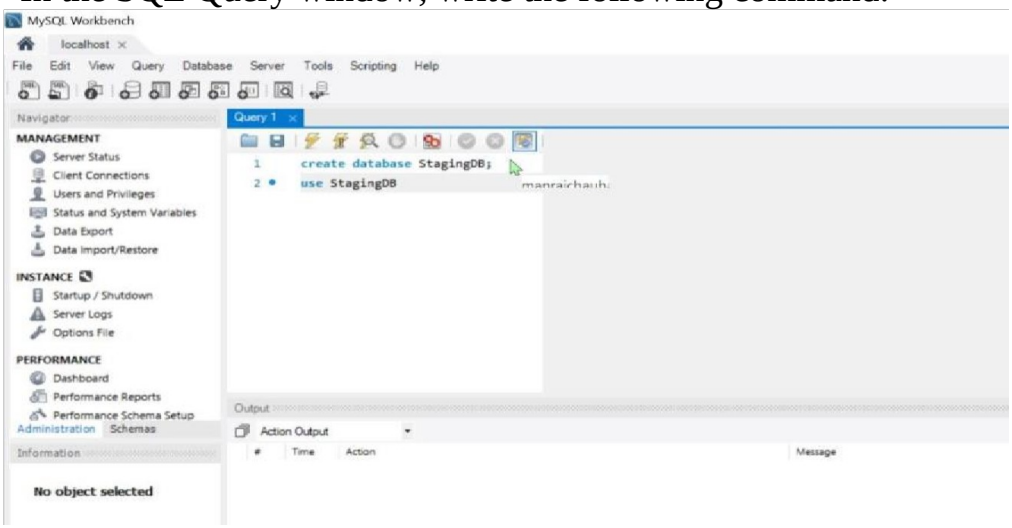
- Hostname: 127.0.0.1
- Port: 3306
- User: root Enter your password and click **OK**





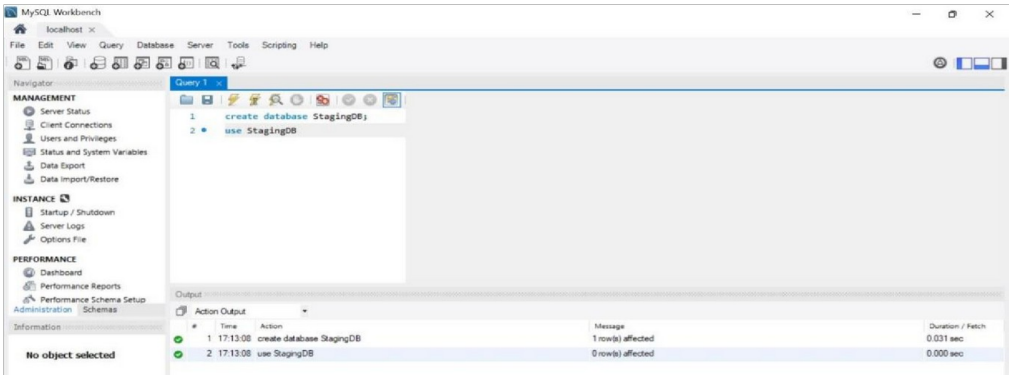
Step 2: Create a New Database (Schema)

In the SQL Query window, write the following command:



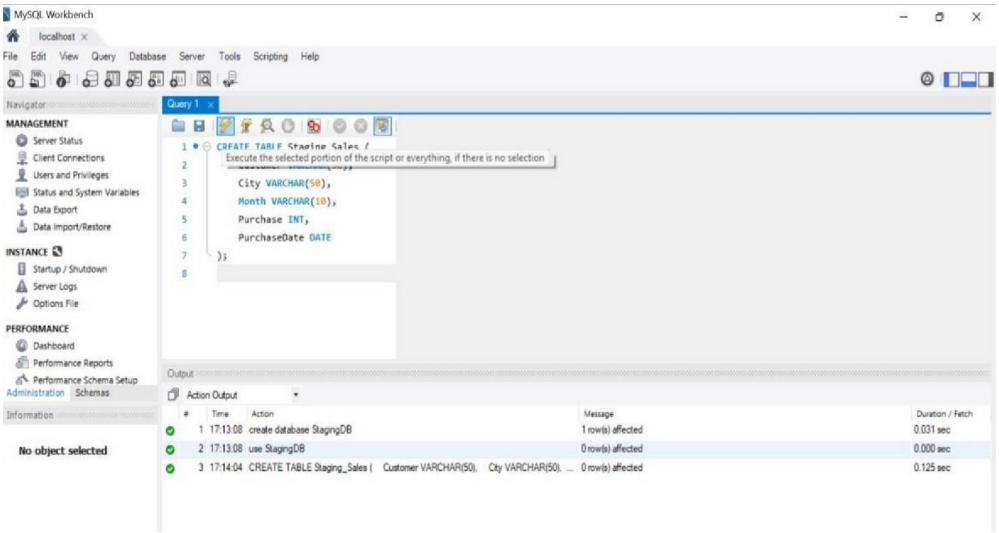
Step 3: Select the Database

Choose the database for further operations:



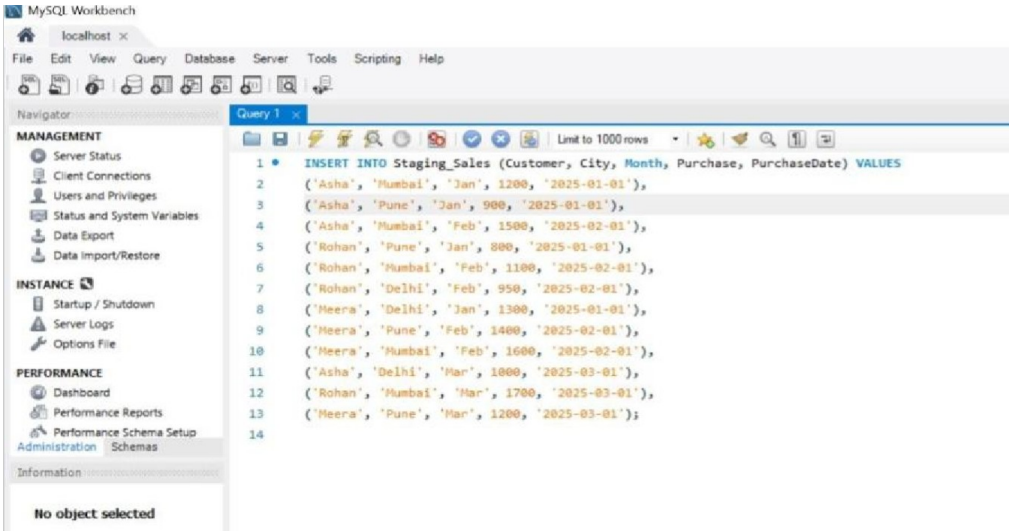
Step 4: Create a Staging Table

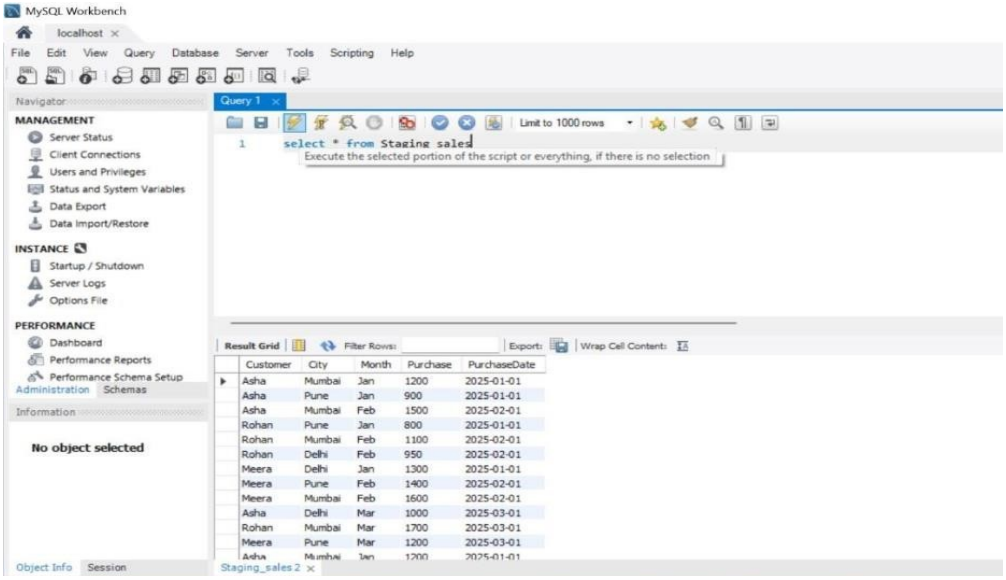
Create a table to store customer sales data:



Step 5: Insert Sales Data into the Staging Table

Insert the given records into staging table:



Step 6: View the Staging Table Data

To display the inserted data: